Axiom at DialectSentEval 2026: Dialect-Aware Ensembling of MARBERTv2 and AraBERTv2 for Arabic Sentiment Analysis

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Abstract

Sentiment analysis of regional Arabic dialects presents a formidable challenge due to severe orthographic variation, morphological complexity, and a scarcity of annotated resources. To tackle these linguistic hurdles within Subtask-1 of the DialectSentEval 2026 Shared Task, team **Axiom** proposes a heavily regularized, dialect-aware ensemble framework combining two domain-complementary language models: `MARBERTv2` and `AraBERTv2`. Our pipeline incorporates preprocessed dialect tags appended directly to input sequences, Stochastic Weight Averaging (SWA) for flat-minima convergence, and simple probability averaging across model outputs. Empirical results on the hidden test set demonstrate the robustness of this approach, with our system securing **2nd place** overall and achieving an official **Macro F1-Score of 0.8211**.

1 Introduction

Arabic sentiment analysis has witnessed significant progress with the development of transformer-based Pretrained Language Models (PLMs). However, transferring these models to regional Arabic dialects remains fundamentally challenging. Unlike Modern Standard Arabic (MSA), dialects exhibit diglossia, severe orthographic ambiguity, rich but non-standardized morphology, and frequent code-switching (Farghaly and Shaalan, 2009; Hamed et al., 2025; Alayba, 2025). Because dialectal Arabic is primarily spoken and lacks standardized orthography, the resulting token sparsity and surface-form variations severely degrade the performance of models trained exclusively on formal MSA (Alhafni et al., 2024; Asadi and Asli-Badarneh, 2026).

The DialectSentEval 2026 shared task provides a critical benchmark for evaluating sentiment analysis systems across diverse Arabic dialects (Ezzini et al., 2026a,b). Building upon foundational dialectal datasets, benchmarks, and shared task initiatives (Alharbi et al., 2025c,a,b), Subtask-1 focuses on 3-class sentiment classification (Negative, Neutral, Positive) over dialectal text sequences.

In this work, we present team **Axiom**'s submission for Subtask-1. We take the stance that when modeling noisy, moderately sized dialectal datasets, optimization stability and domain complementarity are vastly more effective than scaling model capacity or relying on unstable meta-learning stacks. To this end, we propose a highly regularized, zero-added-parameter dual-model ensemble combining `MARBERTv2` (Abdul-Mageed et al., 2021) and `AraBERTv2` (Antoun et al., 2020). By coupling explicit dialect token prefixing with Stochastic Weight Averaging (SWA) (Izmailov et al., 2018; Lu et al., 2022) to find flatter local optima, our pipeline effectively captures both dialectal slang and formal sentiment cues while heavily resisting overfitting. On the official leaderboard, our system achieved **2nd place** overall with an official **Macro F1-Score of 0.8211** and an **Accuracy of 0.8438**.

Our main contributions are summarized as follows:

- The implementation of a simple yet highly effective mechanism that explicitly conditions transformer attention heads on regional metadata to mitigate cross-dialect ambiguity.
- The demonstration that integrating Stochastic Weight Averaging (SWA) into the fine-tuning of Arabic PLMs successfully traverses sharp local minima, stabilizing convergence on highly variable text.
- The construction of a robust probability ensemble that seamlessly fuses the informal semantic representations of MARBERTv2 with the structural MSA bias of AraBERTv2, delivering superior generalization without the overhead of stacked architectures.

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 1419–1425
October 28-29, 2026 ©2026 Association for Computational Linguistics

The remainder of this paper is organized as follows: Section 2 reviews the linguistic challenges and modeling strategies in Arabic dialect NLP. Section 3 outlines our preprocessing, base architectures, and optimization framework. Section 4 presents the experimental setup, official leaderboard results, and ablation analysis. Finally, Section 5 concludes the paper.

2 Related Work

2.1 Linguistic and Orthographic Challenges

Dialectal Arabic sentiment classification is computationally more challenging than MSA processing due to structural diglossia and non-standardized spelling conventions (Boulesnam and Boucetti, 2025; Jaleel, 2025). Dialects differ from MSA across lexical, morphological, and syntactic dimensions (Inoue et al., 2021b; Joubran-Awadie and Shalhoub-Awwad, 2023). Because social media users write dialects phonetically without fixed orthographic standards, token-based NLP models suffer from severe data sparsity (Moaiad et al., 2024; Alnajjar and Hämäläinen, 2024). Consequently, sentiment-bearing adjectives, idioms, and negation patterns frequently appear in forms unobserved by MSA-trained models (Alshemali, 2026; Khalak et al., 2026), underscoring the necessity of dialect-aware normalization and modeling frameworks (Iwidat and Helou, 2026).

2.2 Base Model Selection and Cross-Dialect Transfer

Empirical evidence in Arabic PLM transfer demonstrates that variant proximity between pretraining data and downstream task variety often outweighs raw model size (Inoue et al., 2021a). For informal social media text, `MARBERT` pretrained on a massive Arabic Twitter corpus regularly outperforms general MSA models on dialectal sentiment benchmarks (Abdul-Mageed et al., 2021; Nwesri et al., 2025). Conversely, MSA-pretrained encoders such as `AraBERT` retain superior syntactic and structural representations that generalize remarkably well to unseen dialects and formal expressions (Antoun et al., 2020; Boudad et al., 2023; Muller et al., 2020). Combining these distinct model biases provides a balanced coverage of both informal dialectal expressions and structured sentiment syntax.

2.3 SWA and Ensembling in Low-Resource NLP

Fine-tuning transformer models on small datasets risks converging into sharp, non-generalizable local minima. Stochastic Weight Averaging (SWA) addresses this by averaging SGD trajectory weights during training, effectively guiding the parameters toward wider, flatter regions of the loss landscape that demonstrate superior out-of-distribution generalization (Izmailov et al., 2018; Lu et al., 2022; Demir et al., 2024). Furthermore, ensembling diverse PLMs provides proven gains in uncertainty estimation and predictive robustness (Arango et al., 2024). Prior works in dialectal sentiment analysis confirm that multi-model transformer ensembles consistently outperform single-checkpoint baselines (Mansour et al., 2025; Boughareb et al., 2025), validating our choice of probability-averaged ensembling under tight dataset constraints.

3 Methodology

To address the challenges of dialectal variation and the risk of overfitting on a moderately sized dataset, we eschewed complex meta-learning architectures in favor of a highly regularized, dialect-aware ensemble. Our pipeline consists of four primary stages: text preprocessing, base model selection, fine-tuning via Stochastic Weight Averaging (SWA), and probability-based ensembling.

The high-level data flow of this pipeline is illustrated in Figure 1. As depicted, raw dialectal input is first normalized and augmented with regional tags before branching into two parallel, domain-specific transformer encoders, ultimately fusing their regularized probabilistic outputs to produce the final sentiment classification.

3.1 Preprocessing and Representation

Standard Modern Standard Arabic (MSA) preprocessing often aggressively filters characters, which can inadvertently destroy crucial dialectal markers. We utilized the `ArabertPreprocessor` to clean the text while preserving essential morpho-syntactic boundaries.

To explicitly condition the transformer architectures on regional dialects, we appended a specific dialect token as a prefix to the input sequence. The final input representation is formulated as:

$$x_{\text{input}} = \text{`['} \parallel \text{dialect} \parallel \text{`]'} \parallel \text{preprocess}(s) \quad (1)$$

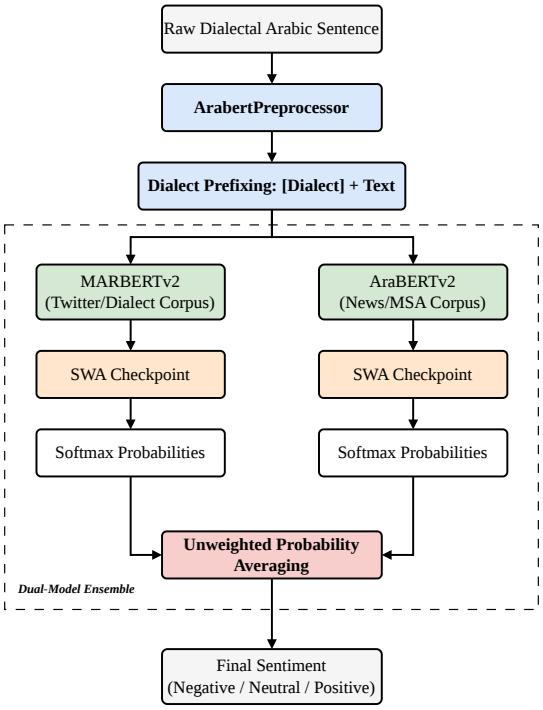


Figure 1: Overall architecture of the Axiom dialect-aware sentiment analysis pipeline.

where s denotes the raw input sentence and $\|$ denotes string concatenation. This prefixing mechanism acts as a soft prompt, guiding the attention heads to contextualize subsequent vocabulary based on the indicated regional dialect.

3.2 Base Architectures

We selected two robust, domain-complementary Transformer encoders to ensure representational diversity within our ensemble:

- **MARBERTv2** (Abdul-Mageed et al., 2021): Pretrained heavily on a massive corpus of Arabic tweets, providing strong zero-shot comprehension of informal phrasing, missing diacritics, and regional slang.
- **AraBERTv2** (Antoun et al., 2020): Pretrained predominantly on news corpora, offering a rigid understanding of structural MSA and formal sentiment expressions that often anchor longer sequences.

3.3 Fine-Tuning and Optimization

To prevent data leakage, we partitioned the training data into a 90/10 stratified split based on the 3-class sentiment labels. Both models were fine-tuned independently using standard Cross-Entropy loss, as margin-based losses (e.g., Focal Loss) were empirically found to overfit on noisy labels in the training set.

To smooth the loss landscape and prevent convergence into sharp local minima, we applied Stochastic Weight Averaging (SWA) (Izmailov et al., 2018). Complete hyperparameter configurations, including learning rate schedules, SWA transition dynamics, and batch statistics, are detailed in Appendix A.

3.4 Ensembling Strategy

For the final predictive framework, we relied on unweighted probability averaging rather than a stacked meta-learner. The output logits from the SWA checkpoints of both models were converted into probability distributions via a Softmax layer. The final sentiment probability was calculated as:

$$P_{\text{ens}}(y|x) = \frac{1}{2}\left[P_{\text{MAR}}(y|x) + P_{\text{Ara}}(y|x)\right] \quad (2)$$

The final predicted class $\hat{y}$ corresponds to $\text{argmax}_y P_{\text{ens}}(y|x)$.

4 Experiments and Results

4.1 Experimental Setup

All models were trained using PyTorch with Automatic Mixed Precision (AMP) on an NVIDIA GPU environment. To facilitate exact replication, comprehensive hardware details, sequence lengths, and optimization configurations are provided in Appendix A.

4.2 Local Validation

During local evaluation on our 10% stratified hold-out set ($N = 174$), both models demonstrated strong convergence. `MARBERTv2` achieved a standalone Macro F1 of 0.9410, while `AraBERTv2` achieved 0.8962. By ensembling both models via probability averaging, the local Macro F1 score peaked at **0.9560**, demonstrating the complementary nature of the two architectures.

4.3 Official Shared Task Results

Our proposed ensemble was evaluated blindly on the official CodaLab test set for Subtask-1. The **Axiom** system achieved **2nd place** overall, recording an official **Macro F1-Score of 0.8211**, an Accuracy of 0.8438, Precision of 0.8188, and Recall of 0.8242.

For context within the competition overview, the first-place system achieved a Macro F1 of 0.8667,

while our ensemble comfortably outperformed the third-place baseline (Macro F1: 0.8175) and the rest of the participating field. These results confirm that our dialect-aware preprocessing paired with flat-minima optimization successfully generalized to unseen dialectal texts, securing a top-tier rank without relying on external datasets or complex meta-learners.

4.4 Ablation and Analysis

To quantify the contribution of our architectural decisions, we tracked the performance of individual components on the official CodaLab blind test set. Table 1 illustrates how our ensembling strategy compares to standalone models and alternative stacking methods. A comprehensive breakdown of all secondary metrics (Accuracy, Precision, Recall) is available in Appendix B.

Model Configuration	Macro F1
MARBERTv2 (Standalone)	0.8096
AraBERTv2 (Standalone)	0.7646
CAMeLBERT-DA (Standalone)	0.7656
Dual-Model Ensemble (Ours)	**0.8211**
Ensemble + LR Meta-Learner	0.8134
3-Model Ensemble (+ CAMeLBERT)	0.7932

Table 1: Summary of official test scores. Detailed metrics are provided in Appendix B.

Base Model Performance: In isolation, MARBERTv2 proved to be the strongest standalone architecture (0.8096 F1), validating the hypothesis that pretraining on social media text aligns better with the noisy dialectal target domain. Conversely, AraBERTv2 and CAMeLBERT-DA struggled to generalize when acting alone.

The Power of Complementary Ensembling: Despite AraBERTv2's weaker standalone performance, fusing its probabilistic outputs with MARBERTv2 yielded our highest official score, culminating in an F1-Score of **0.8211**. This confirms that AraBERT captures distinct structural or formal sentiment cues that MARBERT misses, making them highly complementary.

Detrimental Additions: Throughout development, we observed that adding complexity often harmed generalization. Replacing our simple probability average with a Logistic Regression stacking meta-learner degraded the F1 score to 0.8134, likely due to the meta-learner overfitting the highly confident out-of-fold predictions from the small training set. Similarly, introducing CAMeLBERT-DA as a third model introduced destructive noise. This indicates that misaligned pretraining vocabularies can actively harm small-scale probability ensembles, reinforcing the necessity of strict model curation.

4.5 Error Analysis

While our ensemble generalized exceptionally well locally, a qualitative review of the misclassified validation samples revealed three primary failure modes driven by linguistic complexity. **(1) Subjective Neutrality:** Texts containing implicit complaints alongside neutral anchors were often aggressively predicted as *Negative* by both models, despite being ground-truthed as *Neutral*. **(2) Conflicting Polarities:** Sentences balancing negative aspects with positive justifications confused the models, causing the ensemble to default to *Neutral*. **(3) Negated Negativity:** Formal phrases implying a positive experience through the negation of a negative were conservatively misclassified as *Neutral*. A detailed breakdown with specific Arabic text examples is provided in Appendix C.

5 Conclusion

In this paper, we described team Axiom's 2nd-place submission for Subtask-1 of the DialectSentEval 2026 Shared Task. We demonstrated that for highly dialectal Arabic sentiment analysis on constrained datasets, a heavily regularized dual-model ensemble consisting of MARBERTv2 and AraBERTv2 yields highly competitive results. By combining explicit dialect tagging with Stochastic Weight Averaging and unweighted probability averaging, our system effectively mitigated overfitting while maximizing representational diversity between social media and formal MSA distributions. Future work will explore the integration of generative data augmentation or adversarial training methods to further close the performance gap on rare dialectal structures without degrading model calibration.

6 Limitations

Despite its strong empirical performance, our approach presents three primary limitations. First, ensembling `MARBERTv2` and `AraBERTv2` inherently doubles inference latency and memory overhead compared to standalone models. Second, our maximum sequence length of 128 tokens risks truncating long-range context in extended dialectal narratives. Finally, our dialect prefixing mechanism relies on gold-standard regional metadata; deploy-

ing this pipeline in the wild would necessitate an upstream dialect identification module, potentially introducing cascading classification errors.

Acknowledgments

We extend our gratitude to the organizers of the DialectSentEval 2026 Shared Task and the Fourth Arabic Natural Language Processing Conference for providing the datasets and fostering collaborative research in Arabic NLP.

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A Implementation and Hyperparameter Details

To facilitate reproducibility, we provide the exact hyperparameters utilized during the training phase. All models were trained utilizing a maximum sequence length of 128 tokens and a batch size of 16. We employed the AdamW optimizer with a base learning rate of 2×10^{-5} and a weight decay of 0.01. A linear learning rate scheduler with a 10% warmup period was applied during the initial phase.

Stochastic Weight Averaging (SWA) (Izmailov et al., 2018) was activated at epoch 3, transitioning to an SWA-specific learning rate of 1×10^{-5}. The SWA weights were updated according to the following schedule:

$$w_{\text{SWA}} \leftarrow \frac{w_{\text{SWA}} \cdot n_{\text{models}} + w}{n_{\text{models}} + 1} \tag{3}$$

Batch Normalization statistics were updated across the entire training loader at the conclusion of the 10th epoch. All training was conducted using PyTorch with Automatic Mixed Precision (AMP) to optimize VRAM utilization on an NVIDIA GPU.

B Detailed Ablation Metrics

Table 2 provides the comprehensive evaluation metrics (Accuracy, Precision, Recall, and Macro F1) for all tested configurations on the official blind test set.

C Qualitative Error Analysis Examples

Table 3 provides specific examples from our local 10% validation holdout where the dual-model ensemble failed to predict the correct sentiment. These examples illustrate the primary failure modes discussed in Section 4.5, specifically highlighting the difficulty of resolving conflicting polarities and implicit sentiment in dialectal Arabic.

Model Configuration	Accuracy	Precision	Recall	Macro F1
Standalone Base Models:				
MARBERTv2	0.8324	0.8067	0.8138	0.8096
CAMeLBERT-DA	0.7829	0.7667	0.7765	0.7656
AraBERTv2	0.7886	0.7652	0.7700	0.7646
Dual-Model Ensemble (Ours)	**0.8438**	**0.8188**	**0.8242**	**0.8211**
Alternative Formulations:				
Dual-Ensemble + LR Meta-Learner	0.8305	0.8121	0.8207	0.8134
3-Model Ensemble (+ CAMeLBERT)	0.8133	0.7920	0.8012	0.7932

Table 2: Comprehensive ablation study of official test scores, demonstrating the impact of base model selection and various ensembling strategies.

Raw Arabic Input & Translation	True	Pred.	Failure Mode
متوسط، ممكن يكون أحسن و توقعنا ان الغرف والحمامات تكون أنظف قضينا يومين من ليالينا الأولى في القاهرة هناك *(Average, could be better and we expected the rooms and bathrooms to be cleaner...)*	Neutral	Negative	Subjective Neutrality / Implicit Complaint
بالرغم من بلي فندق كايعطي شي ثمن كبيرا مقارنة بالجيران، و لكن أنا كنت بيه عشان كان مشهور و كان عندو تقييمات. *(Even though the hotel charges a high price compared to neighbors, I stayed because it was famous...)*	Negative	Neutral	Conflicting Polarities (Mixed Sentiment)
وصلنا هناك قبل يوم من الموعد المحدد، لكن هذا لا مشكلة في فهمي. *(We arrived there a day before the scheduled time, but this is no problem...)*	Positive	Neutral	Negated Negativity (Implicit Positivity)

Table 3: A representative sample of misclassified instances from the local validation set, highlighting the challenges of implicit phrasing and boundary ambiguity in dialectal Arabic.